Formulating Camera-Adaptive Color Constancy as a Few-shot Meta-Learning Problem

Steven McDonagh*, Sarah Parisot*, Fengwei Zhou, Xing Zhang,
Ales Leonardis, Zhenguo Li, Gregory Slabaugh
Huawei Noah's Ark Lab
{steven.mcdonagh, sarah.parisot}@huawei.com

1. Introduction

Digital camera pipelines employ color constancy methods to estimate an unknown scene illuminant, enabling the generation of canonical images under an achromatic light source. By taking advantage of large amounts of labelled images, learning-based color constancy methods provide state-of-the-art performance. However, resulting models typically suffer from domain gaps and fail to generalise across imaging devices. Data collection is typically arduous, as it requires both imaging physical calibration objects across different settings (such as indoor and outdoor scenes), as well as manual image annotation to produce ground truth labels. Few works have attempted to mitigate the costs of sensor-specific data collection, calibration and image labelling. Recent unsupervised approaches [1] yield subpar performance compared to fully-supervised methods, while attempts to learn a transform between Camera Spectral Sensitivity (CSS) [6] require a priori knowledge of the CSS and is limited to pairs of sensors.

In this work, we propose a new approach that affords fast adaptation to previously unseen cameras, and robustness to changes in capture device by leveraging annotated samples across different cameras and datasets. We present a general approach that utilizes the concept of color temperature to frame color constancy as a set of distinct, homogeneous few-shot regression tasks, each associated with an intuitive physical meaning. We integrate this novel formulation within a meta-learning framework, enabling fast generalisation to previously unseen cameras using only handfuls of camera specific training samples. Consequently, the time spent for data collection and annotation substantially diminishes in practice whenever a new sensor is used. We evaluate our pipeline on three publicly available datasets comprising 12 different cameras and diverse scene content. Our approach shows a clear improvement over a standard fine-tuned baseline both qualitatively and quantitatively.

2. Camera-Adaptive Color Constancy

An overview of our proposed Meta-AWB method is shown in Fig. 1. We consider a set of datasets $\Delta = \{\mathcal{D}_j\}$, where each dataset $\mathcal{D}_j = \{C_j, \{I_i\}_j\}$ comprises images acquired using a single camera C_j, representing various scenes under varying illumination conditions. Our objective is to provide a color constancy framework that is robust to variations in capture device, so as to leverage all the data available in Δ in order to adapt to previously unseen cameras with limited new training samples. We propose a physically intuitive model that casts the color constancy problem as a set of simple illuminant and camera specific regression tasks. We use the concept of color temperature to approximate the type of light source illuminating each image in our datasets, allowing us to separate images based on light source in an unsupervised manner. As tasks are illuminant specific, estimated illuminant corrections have limited variability and can be evaluated accurately with limited training samples. This allows us to frame each task as a few-shot regression problem, which can be addressed using recent few-shot learning strategies, such as meta-learning.

Color Temperature (CT) is a common measurement in photography, often used in high-end camera software to describe the color of the illuminant for setting white balance [9]. By definition CT measures, in degrees Kelvin, the temperature that is required to heat a Planckian radiator (defined as a theoretical object that is a perfect radiator of visible light [11]) to produce light of a particular color. The Planckian locus, is the path that the color of a Planckian radiator would take in chromaticity space as the temperature increases, effectively illustrating all possible CTs. In practice, the chromaticity of most light sources is off the Planckian locus, so the Correlated Color Temperature (CCT) is computed. CCT is the point on the Planckian locus closest to the non-Planckian light source [11]. Intuitively, CCT describes the color of the light source and can be approximated from photos taken under this light. For each image, we can compute CCT using standard approaches that map

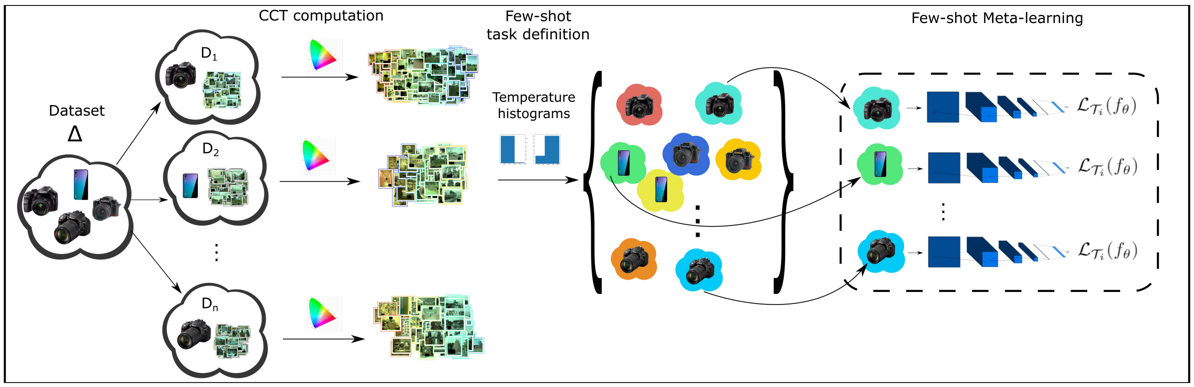


Figure 1: Overview of the proposed Meta-AWB few-shot strategy. Considering a set of cameras and camera specific images, we define a task distribution $\mathcal{T}_i \sim p(\mathcal{T})$ by separating images based on illuminant color. This is done by computing correlated color temperature (CCT) for each image, and building a CCT histogram for each camera. Images in the same task are defined as images captured using the same camera and belonging to the same CCT histogram bin.

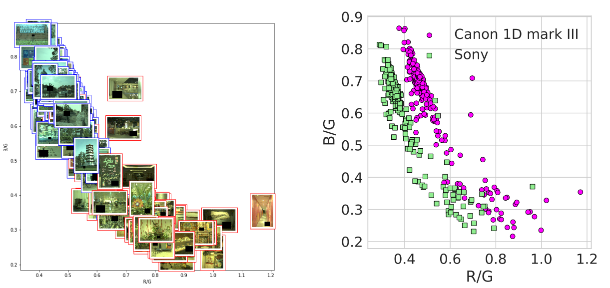


Figure 2: (left) Pre-white balance images, marking their individual ground-truth gain corrections in $[\frac{r}{g}, \frac{b}{g}]$ 2D space. Image frame border color (red, blue) indicates corresponding image temperature histogram bin membership. (right) Ground-truth gain corrections for images observing identical scenes under similar illumination yet captured with distinct cameras (Canon1D, Sony; NUS-9 [2]).

CIE 1931 chromaticities x and y to CCT [8]. Chromaticities x, y are coordinates in the chromaticity space that can be easily estimated from image RGB values [10].

Illuminant and camera-specific tasks. Traditional gamut based color constancy methods [3, 5] assume that the color of the true scene illuminant is constrained by the colors observed in the image. We make a similar hypothesis by regrouping images with similar dominant colors in the same task. We therefore decompose the inter-camera color constancy problem as a set of regression problems that each comprise images acquired *with the same camera and with similar CCT* (i.e. similar dominant color). Our intuition is that, for each of these problems, illuminant corrections are clustered such that good performance can be obtained quickly with only a handful of training samples. This is motivated by the fact that device-specific Camera Spectral Sensitivities (CSS) affect the color domain of captured images and the recording of scene illumination.

We compute a histogram H_s for camera s containing M bins of CCT values and define each task as containing the set of images in each histogram bin. As a result, we define task i; $\mathcal{T}_i(\mathcal{D}_s, m) \in \mathcal{T}$ as: $\mathcal{T}_i(\mathcal{D}_s, m) = \{ I \mid a_s^m \leq CCT(I) \leq b_s^m,\ Cam(I) = C_s \}$ where $Cam(I)$ is the camera used to acquire image I, and a_s^m, b_s^m are the edges of bin m in histogram H_s. Intuitively, images within the same temperature bin will have a similar dominant color, and therefore one could expect them to have similar illuminant corrections. A large variety of common light sources are defined by relatively low temperatures [8]. Accounting for this non-uniform distribution, we define bin edges of H_s as a partition of temperature values on a logarithmic scale. In particular, for $M = 2$, we expect to separate images under a *warm* light source (*eg.* indoor images) from images under a *cold* light source (*eg.* outdoor images). This effect is illustrated in Fig. 2 (left) where images are plotted marking their respective ground-truth gain correction in $[\frac{r}{g}, \frac{b}{g}]$ space and image frame border colors indicate temperature bin membership. This low granularity decomposition yields a few-shot scenario such that only 10 to 20 training images will be required to adapt to a previously unseen camera.

Meta-learning formulation With the task distribution defined, we can then frame camera-adaptive illuminant estimation as a few-shot learning problem where our tasks compose learning episodes. We propose to use meta-learning techniques such as the popular MAML algorithm [4]. MAML is an iterative algorithm that learns a global set of parameters θ^* across tasks by optimising fine-tuning performance on each training task. The objective is to learn an optimal model *initialisation* capable of achieving strong performance on a new unseen task in only a few gradient updates, using only a small number of training samples.

Each regression task instance $\hat{\boldsymbol{\rho}}_\theta = f_\theta(I)$ aims to estimate a global illuminant correction vector $\boldsymbol{\rho} = [r, g, b]$ for an input image I, where f_θ is a nonlinear function described by a neural network model. The model's parameters θ are learned by minimising the commonly used angular error loss between the ground truth and estimated illuminants:

$$\mathcal{L}(\hat{\theta}) = \arccos(\frac{\hat{\boldsymbol{\rho}}_\theta}{\| \hat{\boldsymbol{\rho}}_\theta \|} \cdot \frac{\boldsymbol{\rho}}{\| \boldsymbol{\rho} \|}). \quad (1)$$

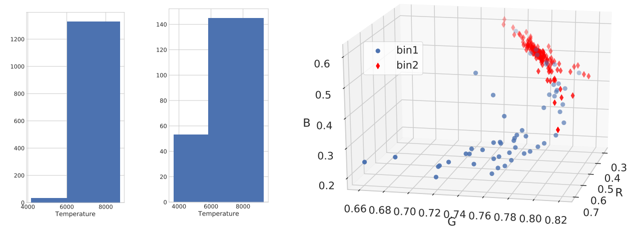
We use the MAML algorithm to train a joint model across all regression tasks. At test time, parameters are fine-tuned for a new unseen task for n gradient updates using K training samples. We finally compute the illuminant correction for each test image I as $\boldsymbol{\rho}_{\theta_i} = f_{\theta_i}(I)$.

3. Results

Datasets and preprocessing. Three public color constancy datasets: Gehler-Shi [12, 7], NUS-9 [2], and Cube [1] are utilised to investigate the capabilities of the proposed methodology. We use a total of 4128 images captured by 12 different cameras. For each dataset, we make use of the provided 'almost-raw' PNG images for all experimental work that follows in Section 3. The **Gehler-Shi** dataset [12, 7] contains 568 images of indoor and outdoor scenes, captured using Canon 1D and Canon 5D cameras. The **NUS-9**-Camera dataset [2] consists of 9 subsets of $\sim$210 images per camera providing a total of $(1736 + 117^1) = 1853$ images. All subsets comprise images representing *the same scene*, highlighting the influence of the camera sensor, illustrated in Fig 2 (right) for two example cameras. The **Cube** dataset [1] contains 1365 images and consists of predominantly outdoor imagery (Canon EOS 550D camera). It was recently updated to include an additional 342 indoor images (renamed **Cube+** dataset). We use all Cube+ data at train time and limit evaluation to the original Cube dataset, in order to be directly comparable to previous work. We apply black-level corrections, a standard gamma correction ($\gamma = 2.2$) and normalize network input to $[0, 1]$. Input images are converted to 8-bit where required, providing bit depth consistency across all datasets.

Implementation. For each camera, we train a model using random image crops of variable size (128×128 to original image size) from the remaining 11 cameras, spatially resized to 128×128. We adopt a simple model architecture comprising four convolutional layers (all sized $3 \times 3 \times 64$), an average pooling layer and two fully connected layers (sizes 64×64, 64×3), with ReLU activations on all hidden layers.

[1]The NUS dataset has recently been updated to include 117 additional images from a ninth camera. During training we use all nine cameras.



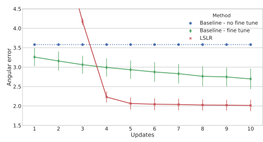
(a) Cube (b) Canon600D (NUS) (c) Canon 600D (NUS) ground-truth illuminants in RGB space

Figure 3: Task definition: Temperature histograms and corresponding separation in RGB space.

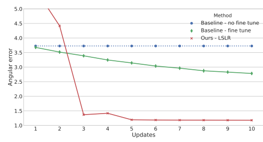
We train our CNN model for $25k$ iterations, using a meta-batch size of 10, number of training images per batch $K_{train} = 10$, learning rate $\beta = 0.001$ (with exponential decay). The MAML inner-update learning rate α is set to 0.001. We use layer normalisation on all convolutional layers. At inference time, for each test image, we randomly select K_{test} training samples (from the test image task) and fine-tune the model for 10 iterations. We independently repeat 10 random draws for each test image and report the median angular error over all images, averaged over all draws. As a baseline, we train our CNN model with standard backprop and leave-one-out cross validation on each camera. At test time we report both with (*Baseline - fine tune*) and without (*Baseline - no fine tune*) K-shot fine tuning. Baselines are trained for $25k$ iterations using the same parameters as our Meta-AWB model.

Histogram-based task definition. To explore the validity of our learning task formulation, we plot CCT histograms per camera (examples in Fig 3(a), 3(b) for M=2 bins) and ground-truth $[r, g, b]$ gain corrections per image in RGB space, with corresponding CCT bin assignment indicated in Fig 3(c). Experimentally we observe high correlation between bin assignment and image colours, content. We find that images can be separated based on light type, and more specifically, indoor vs outdoor light sources. This observation is confirmed in Fig. 3(a) where the CCT histogram, obtained using the original subset of Cube images (all images contain outdoor scenes), essentially contains all images in one bin. Figure 3(c) also illustrates that our CCT-histogram strategy generates homogeneous learning tasks since ground-truth illuminant corrections, belonging to the same bin, are well clustered in RGB space. This setup assigns images to a task, conditioned on both camera sensor and CCT bin, resulting in $M \cdot |\{C_j \in \Delta\}|$ valid learning tasks assuming that each CCT bin is non-empty. We set $M = 2$ in the remaining experiments.

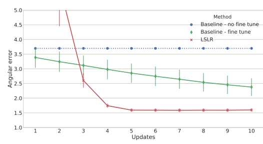
Comparisons to the baseline Figure 4 shows the evolution of the median angular error (lower is better), with



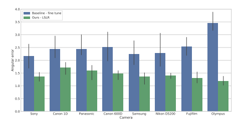
(a) Gehler-Shi



(b) Cube



(c) NUS



(d) NUS, one random draw

Figure 4: Median angular-error with respect to the number of fine-tuning updates. (a-c) Dataset specific results compared to the baselines, (d) Per camera angular-error after 10 updates, all NUS-9 cameras. Error bars in (a-c) report inter-draws variance.

(a) Input image

(b) Ground-truth

(c) Baseline fine-tuned

(d) Ours

Figure 5: An example Gehler-Shi test image. Images are shown in sRGB space and clipped at the 97.5 percentile.

respect to the number of gradient fine-tuning updates, for all datasets under both baselines and our approach. We observe a significant gap in performance for all datasets. Fig. 4(d) shows final per camera angular-error performance and highlights that, unlike the baseline (blue) which particularly struggles for some cameras (*eg.* Olympus camera), our method (green) provides a consistent and better performance on each NUS test camera individually, in addition to average performance. We provide example visual comparison of illuminant inference results in Fig. 5.

4. Conclusion

In this work, we propose a novel formulation of the computational color constancy problem that adapts and generalises quickly to a large variety of camera sensors. We exploit the concept of color temperature to approximate the type of light source from images, so as to decompose the problem into a set of simpler regression tasks, each associated with a camera sensor and type of light source. The simplified nature of the obtained regression tasks allows us to cast color constancy as a few-shot learning problem that we address using meta-learning. Experiments across three benchmark datasets and 12 different camera sensors show improved learning ability over standard fine-tuning, resulting in efficient use of only few training samples. Meta-AWB has the ability to generalise quickly and learns to solve the computational color constancy problem in a camera agnostic fashion. This provides the potential for high accuracy performance as new sensors become available yet mitigates arduous and time-consuming calibration of training imagery, required for fully-supervised approaches.

References

[1] N. Banic and S. Loncaric. Unsupervised learning for color constancy. *CoRR*, abs/1712.00436, 2017. 1, 3

[2] D. Cheng, D. K. Prasad, and M. S. Brown. Illuminant estimation for color constancy: why spatial-domain methods work and the role of the color distribution. *JOSA A*, 31(5):1049–1058, 2014. 2, 3

[3] G. D. Finlayson. Color in perspective. *IEEE transactions on Pattern analysis and Machine Intelligence*, 18(10):1034–1038, 1996. 2

[4] C. Finn, P. Abbeel, and S. Levine. Model-Agnostic Meta-Learning for fast adaptation of deep networks. *arXiv preprint arXiv:1703.03400*, 2017. 2

[5] D. A. Forsyth. A novel algorithm for color constancy. *International Journal of Computer Vision*, 5(1):5–35, 1990. 2

[6] S.-B. Gao, M. Zhang, C.-Y. Li, and Y.-J. Li. Improving color constancy by discounting the variation of camera spectral sensitivity. *JOSA A*, 34(8):1448–1462, 2017. 1

[7] P. V. Gehler, C. Rother, A. Blake, T. Minka, and T. Sharp. Bayesian color constancy revisited. pages 1–8, 2008. 3

[8] J. Hernandez-Andres, R. L. Lee, and J. Romero. Calculating correlated color temperatures across the entire gamut of daylight and skylight chromaticities. *Applied optics*, 38(27):5703–5709, 1999. 2

[9] R. Jacobson, S. Ray, G. G. Attridge, and N. Axford. *Manual of Photography*. Taylor & Francis, 2000. 1

[10] J. Schanda. *Colorimetry: understanding the CIE system*. John Wiley & Sons, 2007. 2

[11] E. F. Schubert. *Light-emitting diodes*. E. Fred Schubert, 2018. 1

[12] L. Shi and B. Funt. Re-processed version of the gehler color constancy dataset of 568 images. *http://www. cs. sfu. ca/~color/data/*, 2000. 3